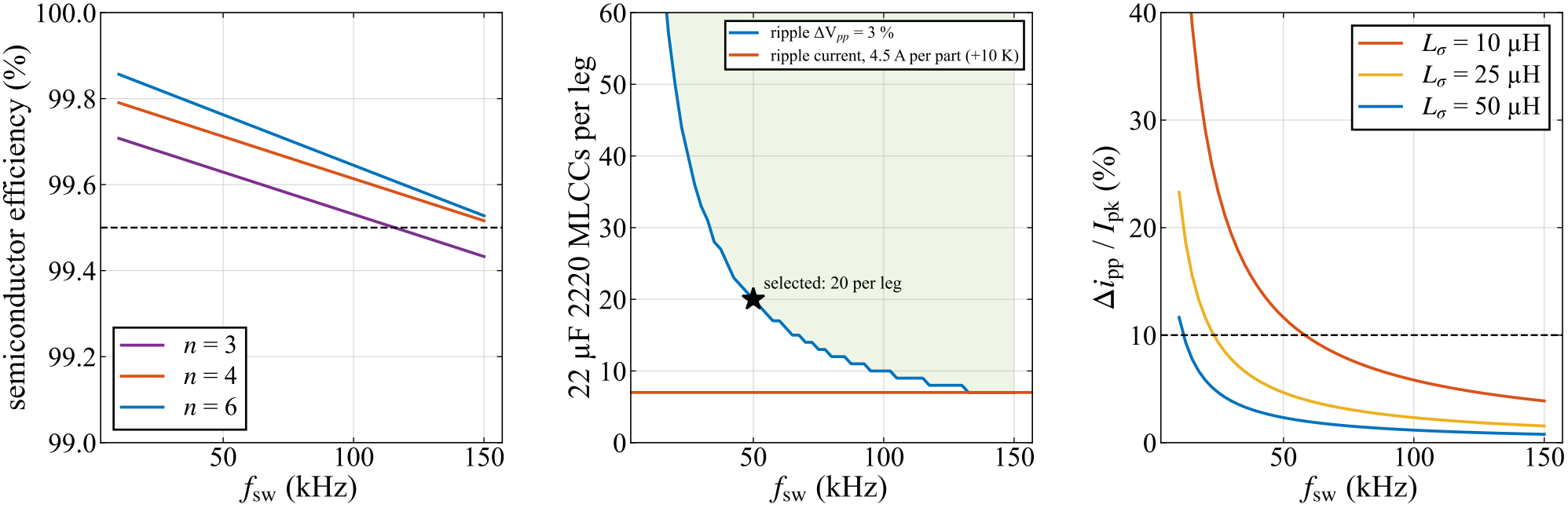
semiconductor efficiency (%)
100.0
99.8
99.6
99.4
99.2
99.0
50
100
150
f_{sw} (kHz)
$n = 3$
$n = 4$
$n = 6$
(a)
22 µF 2220 MLCCs per leg
60
50
40
30
20
10
0
ripple ΔV_{pp} = 3 %
ripple current, 4.5 A per part (+10 K)
selected: 20 per leg
f_{sw} (kHz)
(b)
Δi_{pp} / I_{pk} (%)
40
30
20
10
0
L_σ = 10 µH
L_σ = 25 µH
L_σ = 50 µH
f_{sw} (kHz)
(c)